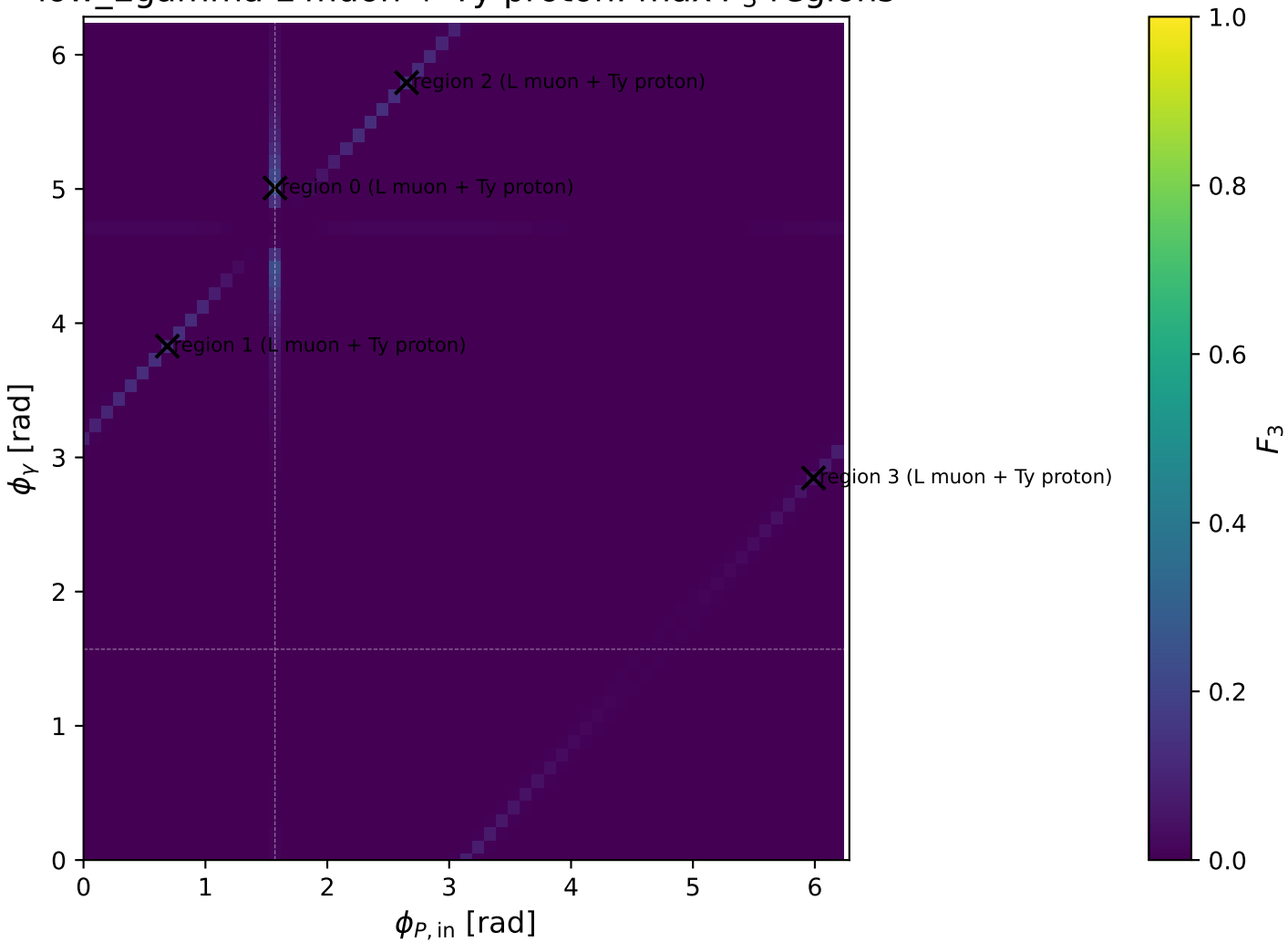
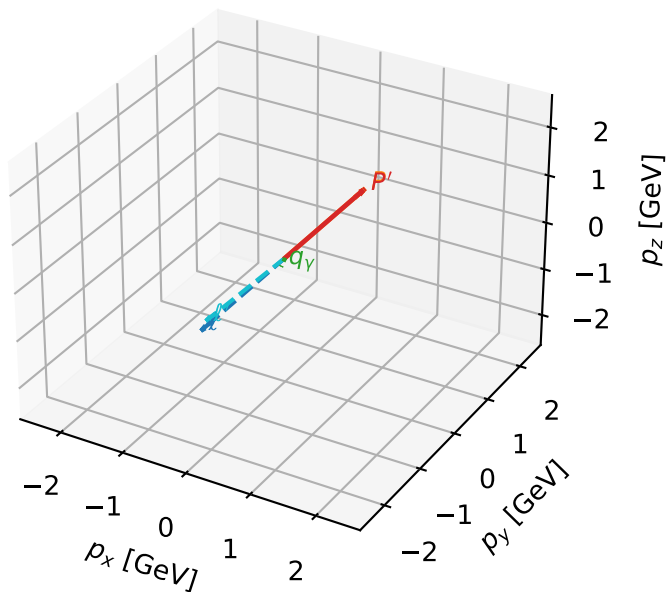


low_Egamma L muon + Ty proton: max F_3 regions



L muon + Ty proton characteristic max F_3 configuration

3D momenta

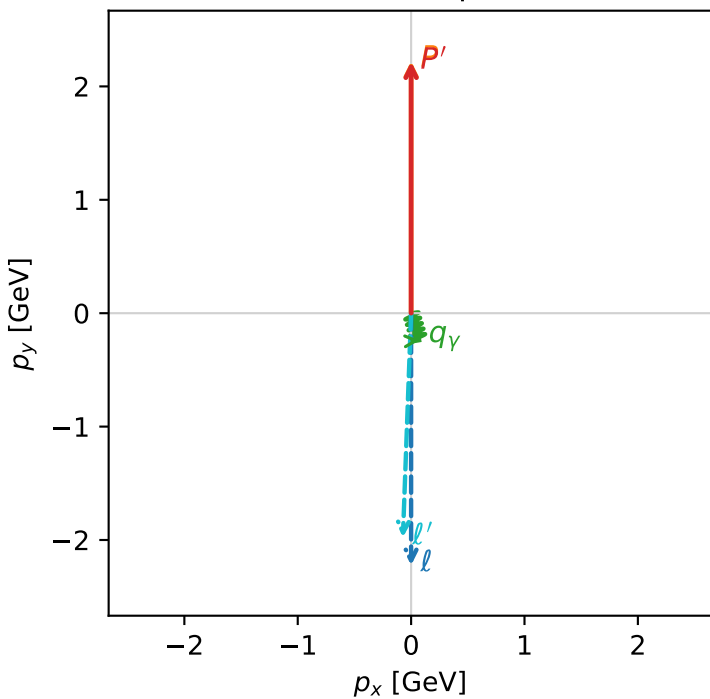


f3_low_egamma_lty_region_0 $F_3=0.230888$
 region: low_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |\bar{T}_{y,p}\rangle$

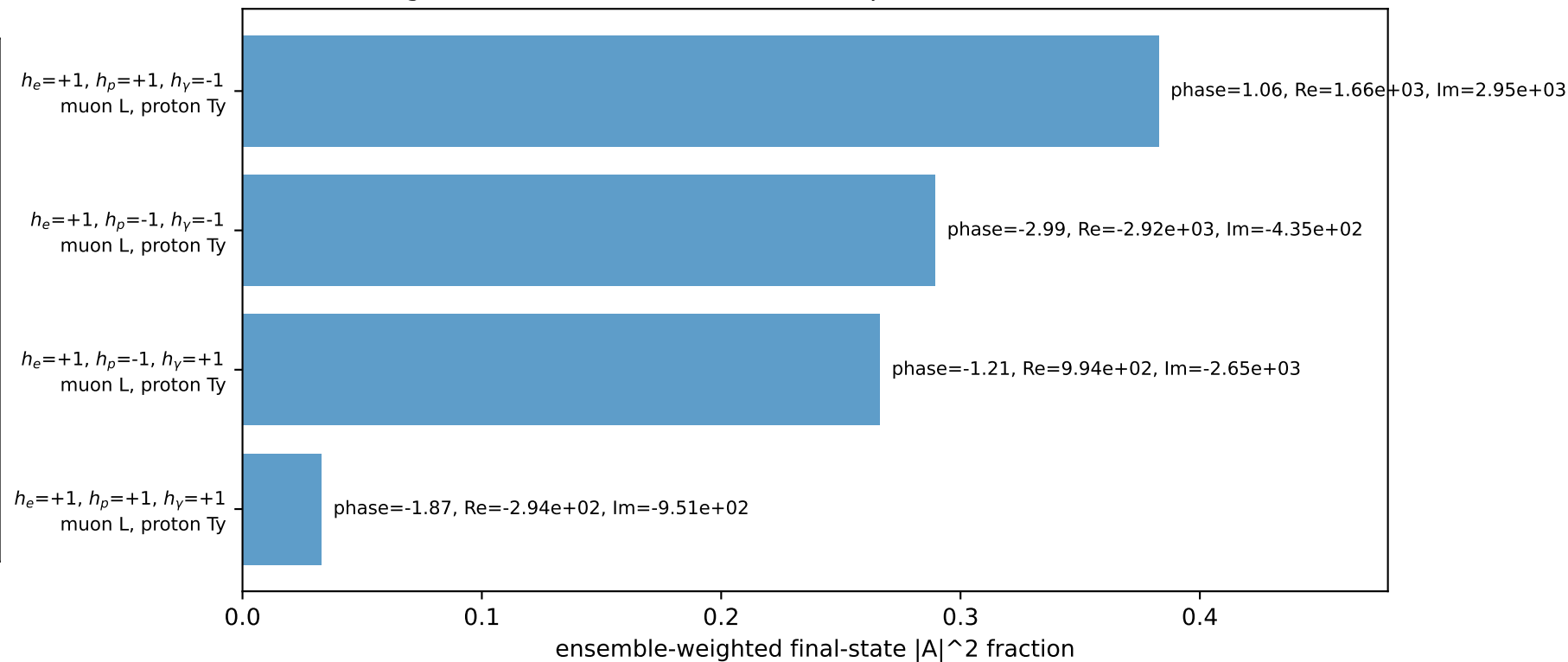
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.21665$
 $\theta_{in}=1.57079$, $\phi_i=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=5.00691$
 $Q^2=0.0060701$, $x_B=0.00267388$, $t=-6.31712e-06$, $W^2=3.14392$, $y=0.110069$
 $\theta(l', \gamma)=18.9768$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 l' $E= 1.9816$ $p_x= -0.0726$ $p_y= -1.9774$ $p_z= 0.0000$
 P' $E= 2.4069$ $p_x= 0.0000$ $p_y= 2.2167$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0726$ $p_y= -0.2392$ $p_z= 0.0000$

Transverse plane

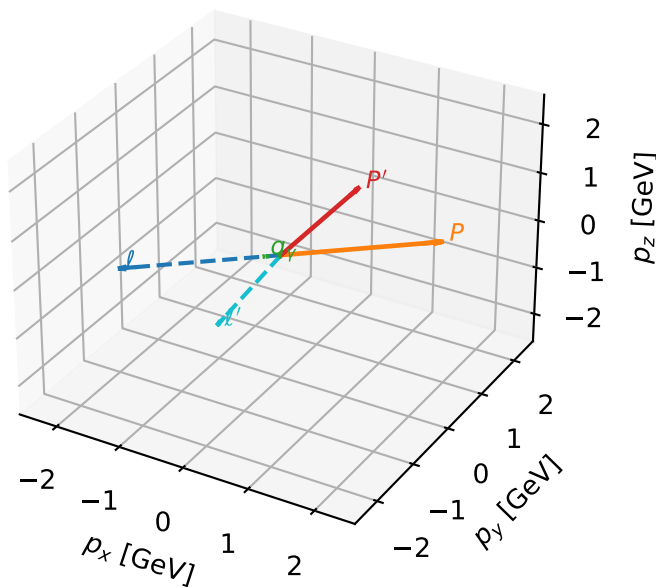


Leading initial-ensemble/final-state components (N=4, retained=97.2%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta

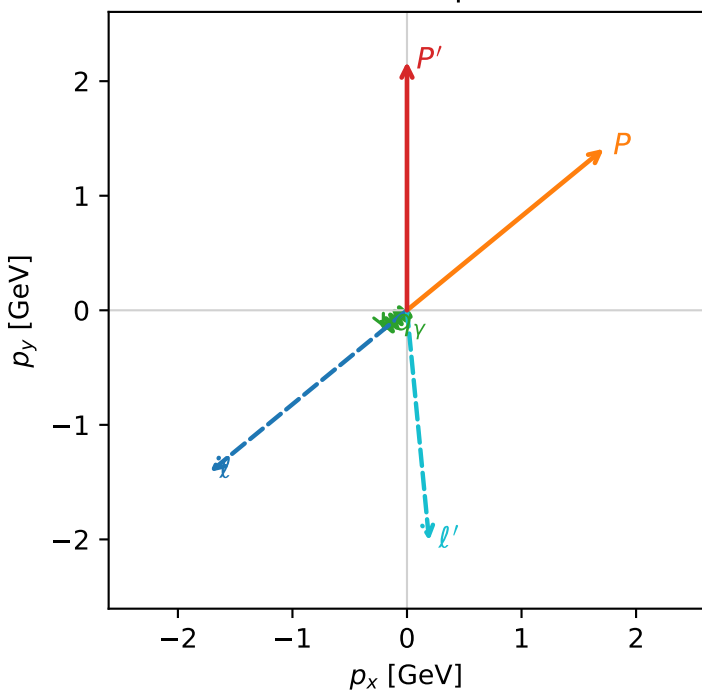


f3_low_egamma_lty_region_1 $F_3=0.133717$
 region: low_Egamma_LTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{Y,p}\rangle$

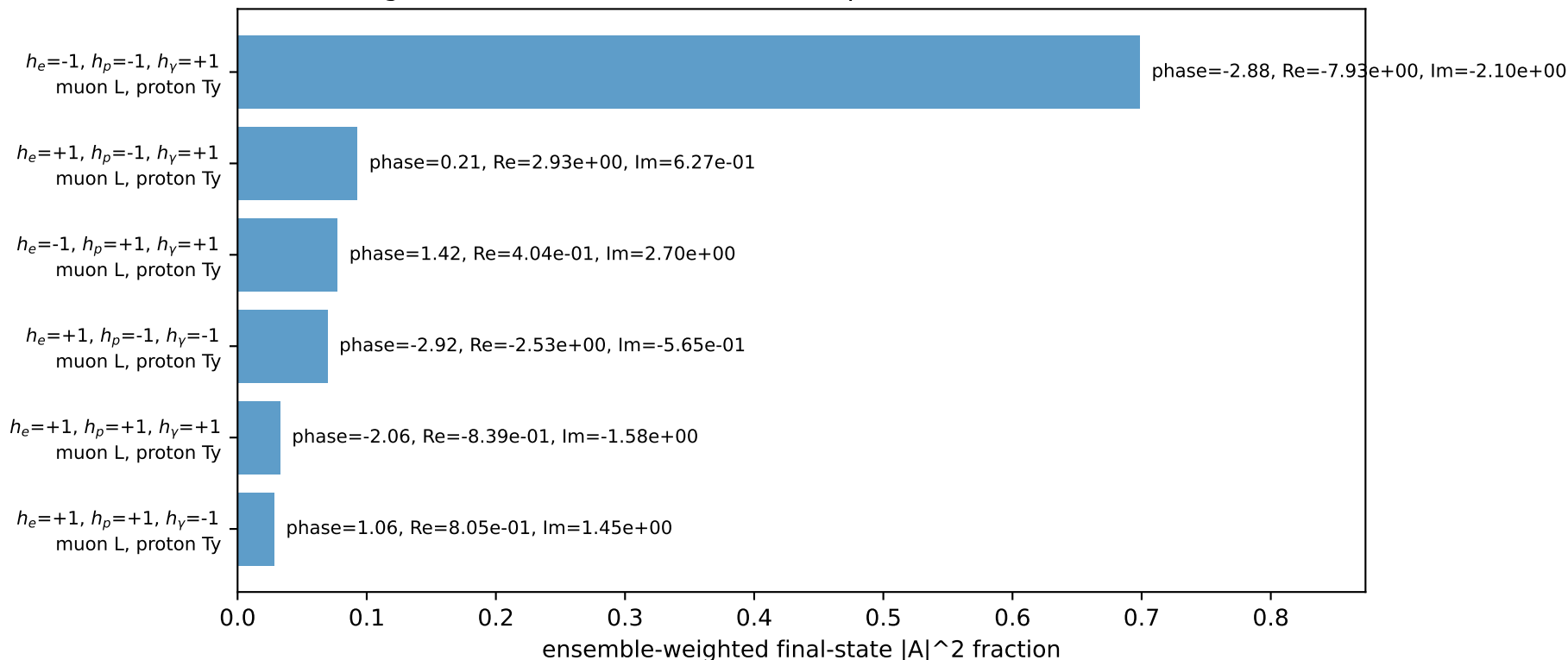
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.17054$
 $\theta_{in}=1.57079$, $\phi_l=3.82882$, $\phi_p=0.687223$
 $E_Y=0.25$, $\phi_Y=3.82882$
 $Q^2=3.97603$, $x_B=0.680036$, $t=-3.5288$, $W^2=2.75061$, $y=0.283483$
 $\theta(l', \gamma)=56.1116$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.7185$ $p_y= -1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.7185$ $p_y= 1.4103$ $p_z= 0.0000$
 l' $E= 2.0240$ $p_x= 0.1933$ $p_y= -2.0119$ $p_z= 0.0000$
 P' $E= 2.3646$ $p_x= 0.0000$ $p_y= 2.1705$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.1933$ $p_y= -0.1586$ $p_z= 0.0000$

Transverse plane

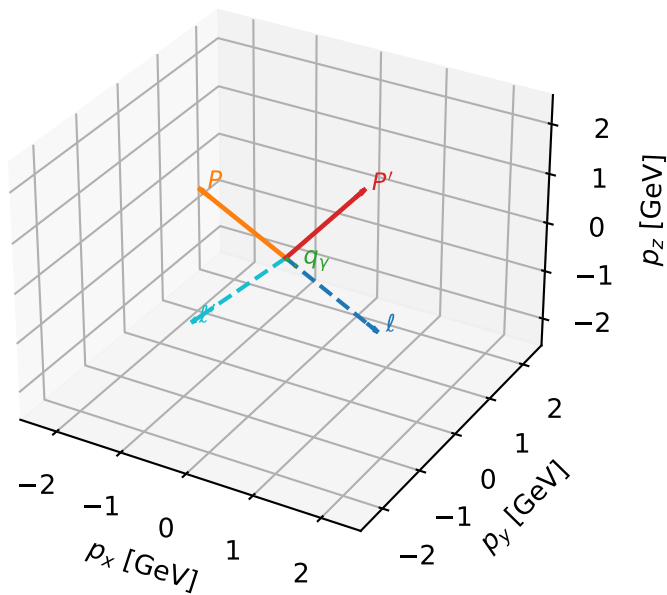


Leading initial-ensemble/final-state components (N=6, retained=100.0%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta



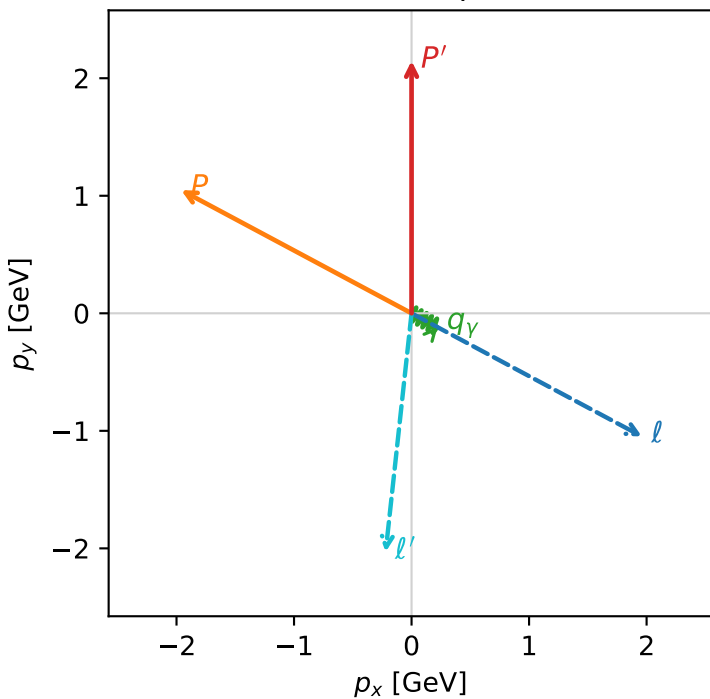
f3_low_egamma_lty_region_2 $F_3=0.126369$
 region: low_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.14791$
 $\theta_{in}=1.57079$, $\phi_i=5.79231$, $\phi_P=2.65072$
 $E_\gamma=0.25$, $\phi_\gamma=5.79231$
 $Q^2=5.68893$, $x_B=0.772209$, $t=-5.04908$, $W^2=2.558$, $y=0.357195$
 $\theta(l', \gamma)=68.0735$ deg

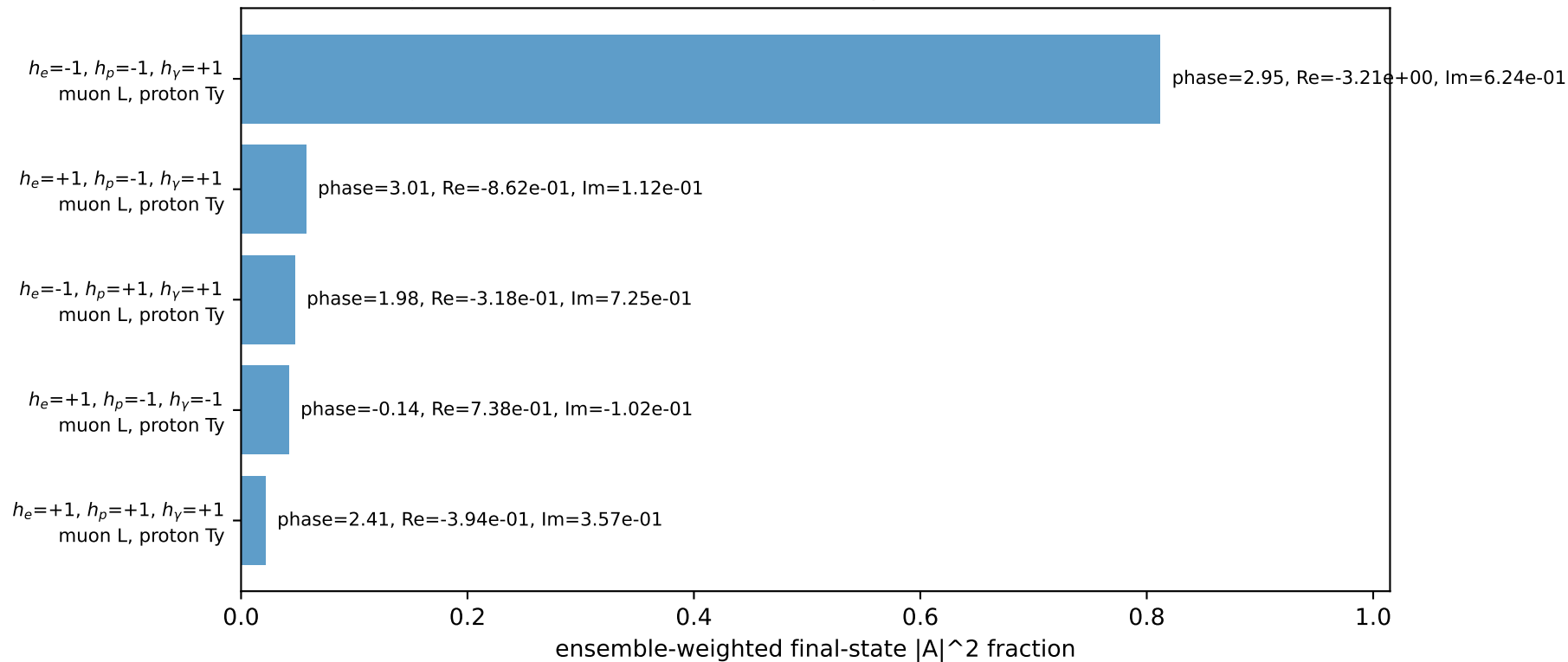
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= 1.9606$ $p_y= -1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.9606$ $p_y= 1.0480$ $p_z= 0.0000$
 l' $E= 2.0447$ $p_x= -0.2205$ $p_y= -2.0301$ $p_z= 0.0000$
 P' $E= 2.3438$ $p_x= 0.0000$ $p_y= 2.1479$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2205$ $p_y= -0.1178$ $p_z= 0.0000$

Transverse plane

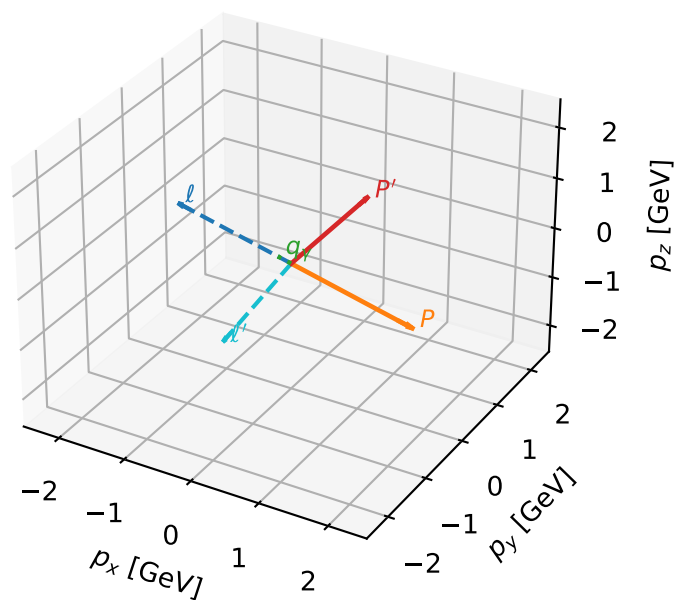


Leading initial-ensemble/final-state components (N=5, retained=98.1%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta

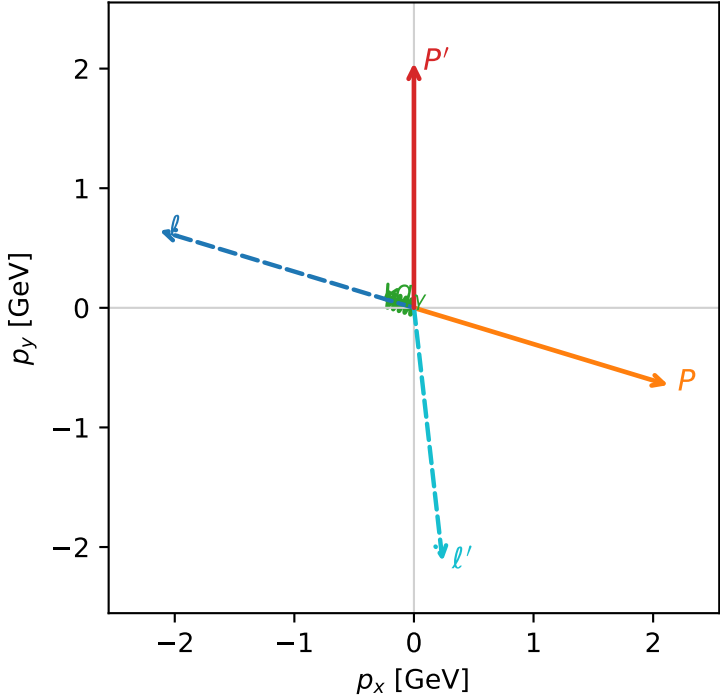


f3_low_egamma_lty_region_3 F_3=0.0599539
region: low_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.04763$
 $\theta_{in}=1.57079$, $\phi_l=2.84707$, $\phi_p=5.98866$
 $E_\gamma=0.25$, $\phi_\gamma=2.84707$
 $Q^2=13.2411$, $x_B=0.941086$, $t=-11.752$, $W^2=1.70877$, $y=0.682188$
 $\theta(l', \gamma)=113.313$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -2.1274$ $p_y= 0.6453$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 2.1274$ $p_y= -0.6453$ $p_z= 0.0000$
 l' $E= 2.1363$ $p_x= 0.2392$ $p_y= -2.1202$ $p_z= 0.0000$
 P' $E= 2.2522$ $p_x= 0.0000$ $p_y= 2.0476$ $p_z= 0.0000$
 q $E= 0.2500$ $p_x= -0.2392$ $p_y= 0.0726$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
muon L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=96.6%)

